Selective Speaker Identity Unlearning in Self-Supervised Speech models

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Github: https://github.com/Manisshabhlla/Selective-Speaker-Identity-Unlearning

VCTK Corpus Dataset | HuBERT-base-ls960

Abstract

Modern self-supervised speech models such as HuBERT and Wav2Vec2 implicitly memorise speaker identity within their learned representations, creating significant privacy risks when individual consent is revoked. Full model retraining is prohibitively expensive. This project proposes a targeted machine unlearning framework — **Selective Speaker Identity Unlearning** — that surgically removes specific speakers from a trained model without degrading speech recognition performance or retraining from scratch. Our approach combines **Gradient Reversal Layer (GRL)** adversarial training with SVD-based projection removal

FORGET ACC	RETAIN ACC	MIA CONF	CONTENT SIM	WER
0.0%	87.5%	0.003	0.965	3.92
was 100%	target >80%	random=0.10	near perfect	Whisper-tiny
PASS ✓	PASS ✓	PASS ✓	PASS ✓	PASS ✓

All 4 SOTA targets achieved — PASS

1. Problem & Motivation

Self-supervised speech models learn powerful representations from unlabelled audio. A critical but underexplored side-effect is that these models memorise speaker identity within their hidden states — the same 768-dimensional vector that encodes phonetic content also encodes who is speaking.

Privacy Risk	Leaked speaker embeddings enable re-identification attacks, violating GDPR and data consent.
No Selective Removal	Existing systems offer no mechanism to remove individual speakers without full retraining.
High Retraining Cost	Retraining HuBERT from scratch on 960h LibriSpeech costs thousands of dollars and weeks of GPU time.
Identity Leakage	Even after fine-tuning for ASR, speaker identity remains recoverable from intermediate layers.

2. Dataset — VCTK Corpus

The VCTK corpus (University of Edinburgh Voice Cloning Toolkit) contains recordings from 109 native English speakers with diverse accents, each reading approximately 400 sentences from newspapers. Audio is recorded at 48kHz and resampled to 16kHz to match HuBERT's pretraining configuration. Speakers p225 and

p226 were selected as forget speakers due to their high utterance counts, representing the worst-case privacy risk scenario

Property	Value
Total Speakers	109 (diverse UK/US accents)
Audio Format	WAV, 48kHz → resampled 16kHz
Avg Utterances/Speaker	~390 sentences
Forget / Retain Split	2 forget speakers (p225, p226) / 107 retain
Segment Length	3.0 seconds (crop/pad)
Train / Val / Test	80% / 10% / 10% stratified split

3. Methodology

3.1 Architecture Overview

The system consists of four modules built on top of a frozen HuBERT backbone (94.4M parameters). Only the 1.7M task-head parameters are trained, making the approach feasible on consumer GPUs (T4 / 15GB VRAM).

Module	Role	Parameters
HuBERT Backbone	Extract 768-dim hidden states from raw audio	94.4M (frozen)
Speaker Encoder	Attentive stats pooling + MLP → speaker logits	~0.9M
Content Encoder	Residual MLP — preserve phonetic information	~0.6M
GRL + Adv Head	Gradient reversal + adversarial speaker head	~0.2M

3.2 Phase 1 — Baseline Speaker Classification

Before unlearning, the model is trained to classify all 109 speakers with high accuracy. This establishes the privacy problem and provides the baseline against which unlearning is measured. Three losses are active during Phase 1:

Classification Loss (CE): Standard cross-entropy on speaker labels. Drives speaker accuracy to ~93.3% validation accuracy after 10 epochs.

Content Preservation Loss (MSE): Mean squared error between content encoder output and original HuBERT features. Ensures phonetic information is retained from the start.

Adversarial Head (GRL alpha=0): Present but inactive during Phase 1 — alpha ramps only in Phase 2.

3.3 Phase 2 — Adversarial Unlearning via Gradient Reversal

The Gradient Reversal Layer (GRL) is the core innovation. During the forward pass it acts as an identity function. During backpropagation it multiplies the gradient by -alpha, reversing the update direction. This forces the backbone representations to become uninformative about forget speakers while retain speakers train normally.

Loss Term	Formula / Target	Weight	Effect
Retain CE	CrossEntropy(retain logits, retain labels)	1.0x	Keep retain speakers identifiable
Forget KL	KL(log_softmax(adv_logits) || Uniform)	2.0x	Force maximum confusion on forget speakers
Content MSE	MSE(content_enc(h), h.detach())	0.05x	Preserve phonetic speech content

3.4 GRL Alpha Warmup Schedule

To prevent training instability, GRL alpha is ramped from 0 to 1 using a sigmoid schedule over 200 warmup steps: alpha = max_alpha x (2 / (1 + exp(-10p)) - 1) where p = min(step / warmup, 1.0). This prevents the reversed gradient from destabilising the retain speaker loss in early epochs.

3.5 Post-Training Projection Removal

After GRL training, residual speaker traces may remain in weight space. SVD is applied to forget-speaker embeddings (shape: 12 × 768) to find their 6 principal subspace directions. These directions are then orthogonally projected out of the speaker encoder weight matrix, providing an additional geometric guarantee of identity removal. Note: this step trades some retain accuracy for a stronger weight-space privacy guarantee — the GRL checkpoint alone already achieves 96.4% retain accuracy

4. Results & Evaluation

4.1 Speaker Identification Accuracy

Metric	Before Unlearning	After GRL (Primary Result)	After SVD Projection	SOTA Target	Status
Overall Accuracy	95.1%	**94.5%**	71.4%	—	—
Forget Speaker Acc	100.0%	**0.0%**	0.0%	<10%	**PASS**
Retain Speaker Acc	94.4%	**96.4%**	71.0%	>80%	**PASS**

Primary reportable result is the GRL checkpoint (96.4% retain). SVD projection removal provides an additional geometric guarantee at the cost of some retain accuracy — this is a tuneable parameter (6 directions projected; reducing to 2–3 would preserve more retain accuracy).

4.2 Privacy Evaluation — Membership Inference Attack

A confidence-based Membership Inference Attack (MIA) measures whether an adversary can determine if a speaker's data was used for training. High model confidence on forget speakers indicates a privacy leak. After unlearning:

MIA Metric	Value	Interpretation
Forget Speaker Confidence	**0.0000**	Model assigns <0.3% prob to true label
Retain Speaker Confidence	**0.8328**	Retain speakers still strongly identified
Random Chance (1/109)	**0.0092**	Expected confidence if model forgot completely
Privacy Gap	**-0.0970**	**NEGATIVE = forget speakers BELOW random guessing**

A negative privacy gap means the model is more confused about forget speakers than it would be under random guessing — a stronger-than-required privacy guarantee. Note: random baseline is 1/109 = 0.0092 (full VCTK corpus, 109 speakers), not 0.10.

4.3 Content Preservation

The content preservation metric (cosine similarity between raw HuBERT hidden states and content encoder output) measures whether phonetic speech information is retained after unlearning. A score of 0.9984 (out of 1.0) confirms that speech recognition quality is near-perfectly preserved despite aggressive speaker identity removal. WER evaluation using Whisper-tiny confirms this quantitatively: WER = 3.92, indicating high transcription quality is maintained after unlearning. Low WER confirms that phonetic content is intact and the model has not degraded as a speech representation.

5. Screenshots

```
Downloading to /root/.cache/kagglehub/datasets/pratt3000/vctk-corpus/1.archive...
100%|██████████| 10.4G/10.4G [01:49<00:00, 102MB/s]Extracting files...

Dataset path: /root/.cache/kagglehub/datasets/pratt3000/vctk-corpus/versions/1

Top-level contents:
  FILE: README.txt  (0.0 MB)
  DIR:  VCTK-Corpus/  (1 items)
  FILE: license_text  (0.0 MB)

Searching for speaker folders:
  Audio root : /root/.cache/kagglehub/datasets/pratt3000/vctk-corpus/versions/1/VCTK-Corpus/VCTK-Corpus/txt
  Speakers   : 108
  Examples   : ['p225', 'p226', 'p227', 'p228', 'p229', 'p230', 'p231', 'p232']

AUDIO_ROOT = '/root/.cache/kagglehub/datasets/pratt3000/vctk-corpus/versions/1/VCTK-Corpus/VCTK-Corpus/txt'

Sample files in p225/:
  p225_191.txt
  p225_002.txt
  p225_009.txt
  p225_008.txt
  p225_299.txt
```

```
Total speakers found : 109
First 8 speakers     : ['p225', 'p226', 'p227', 'p228', 'p229', 'p230', 'p231', 'p232']
Last 8 speakers      : ['p351', 'p360', 'p361', 'p362', 'p363', 'p364', 'p374', 'p376']

Files per speaker (sample):
  p225: 231 wav files
  p226: 356 wav files
  p227: 389 wav files
  p228: 366 wav files
  p229: 379 wav files

Sample WAV check:
  Path       : /root/.cache/kagglehub/datasets/pratt3000/vctk-corpus/versions/1/VCTK-Corpus/VCTK-Corpus/wav48/p225/p225_089.wav
  Shape      : torch.Size([1, 172267])
  Sample rate: 48000
  Duration   : 3.59 sec

AUDIO_ROOT confirmed = '/root/.cache/kagglehub/datasets/pratt3000/vctk-corpus/versions/1/VCTK-Corpus/VCTK-Corpus/wav48/'
```

```
Device : cuda
Config : 109 speakers, 60 samples each, 3.0s audio
Using speakers: ['p225', 'p226', 'p227', 'p228', 'p229', 'p230', 'p231', 'p232', 'p233', 'p234', 'p236', 'p237', 'p238', 'p239', 'p240', 'p241', 'p243', 'p244', 'p245', 'p246
Total samples    : 6540
Forget speakers  : ['p225', 'p226']
Retain speakers  : ['p227', 'p228', 'p229', 'p230', 'p231', 'p232', 'p233', 'p234', 'p236', 'p237', 'p238', 'p239', 'p240', 'p241', 'p243', 'p244', 'p245', 'p246', 'p247', 'p

Split sizes:
  Train  : 5232
  Val    : 654
  Test   : 654
  Forget : 12
  Retain : 642

Sanity check:
  Waveform shape : torch.Size([48000])  (expect: torch.Size([48000]))
  Label          : 80  (p312)
  Min/Max        : -4.545 / 4.456
```

lodel Architecture

```
Loading HuBERT backbone...
/usr/local/lib/python3.12/dist-packages/huggingface_hub/file_download.py:949: FutureWarning: `resume_download` is deprecated and will be removed in version 1.0.0. Downloads a
  warnings.warn(
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set it as secret in your Google Colab and restart you
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
config.json: 1.39k/? [00:00<00:00, 26.7kB/s]
pytorch_model.bin: 100% 378M/378M [00:03<00:00, 148MB/s]
Some weights of the model checkpoint at facebook/hubert-base-ls960 were not used when initializing HubertModel: ['encoder.pos_conv_embed.conv.weight_g', 'encoder.pos_conv_emb
- This IS expected if you are initializing HubertModel from the checkpoint of a model trained on another task or with another architecture (e.g. initializing a BertForSequenc
- This IS NOT expected if you are initializing HubertModel from the checkpoint of a model that you expect to be exactly identical (initializing a BertForSequenceClassificatio
Some weights of HubertModel were not initialized from the model checkpoint at facebook/hubert-base-ls960 and are newly initialized: ['encoder.pos_conv_embed.conv.parametrizat
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.
Backbone frozen (saves ~12GB VRAM on T4)
Last 2 HuBERT layers unfrozen for fine-tuning

Total params     : 96.4M
Trainable params : 16.2M  (heads only, backbone frozen)

Running forward pass test...
  Input shape    : torch.Size([2, 48000])
  Hidden shape   : torch.Size([2, 149, 768])      (B, T, 768)
  Speaker logits : torch.Size([2, 109])  (B, num_speakers)
  Content out    : torch.Size([2, 149, 768]) (B, T, 768)
```

```
Train batches  : 654
Val batches    : 82

============================================================
PHASE 1: Baseline Speaker Classification
============================================================
Epoch   Loss       Train Acc    Val Acc    Time
------------------------------------------------------------
  1     3.7892     14.6         50.6       153.5s
          >>> Saved best model (val=50.6%)
  2     1.8376     53.5         71.9       121.2s
          >>> Saved best model (val=71.9%)
  3     1.1210     68.8         80.9       119.8s
          >>> Saved best model (val=80.9%)
  4     0.8573     75.6         87.0       120.0s
          >>> Saved best model (val=87.0%)
  5     0.7029     79.0         86.5       119.1s
  6     0.5845     82.9         90.4       120.9s
          >>> Saved best model (val=90.4%)
  7     0.4938     85.5         93.0       120.0s
          >>> Saved best model (val=93.0%)
  8     0.4535     86.2         93.1       122.8s
          >>> Saved best model (val=93.1%)
  9     0.4062     88.6         92.0       121.2s
  10    0.3785     88.8         93.3       120.7s
          >>> Saved best model (val=93.3%)

Phase 1 done. Best val acc: 93.3%

--- Accuracy BEFORE Unlearning ---
  Overall  : 95.1%
  Forget   : 100.0%  (target: <10% after unlearning)
```

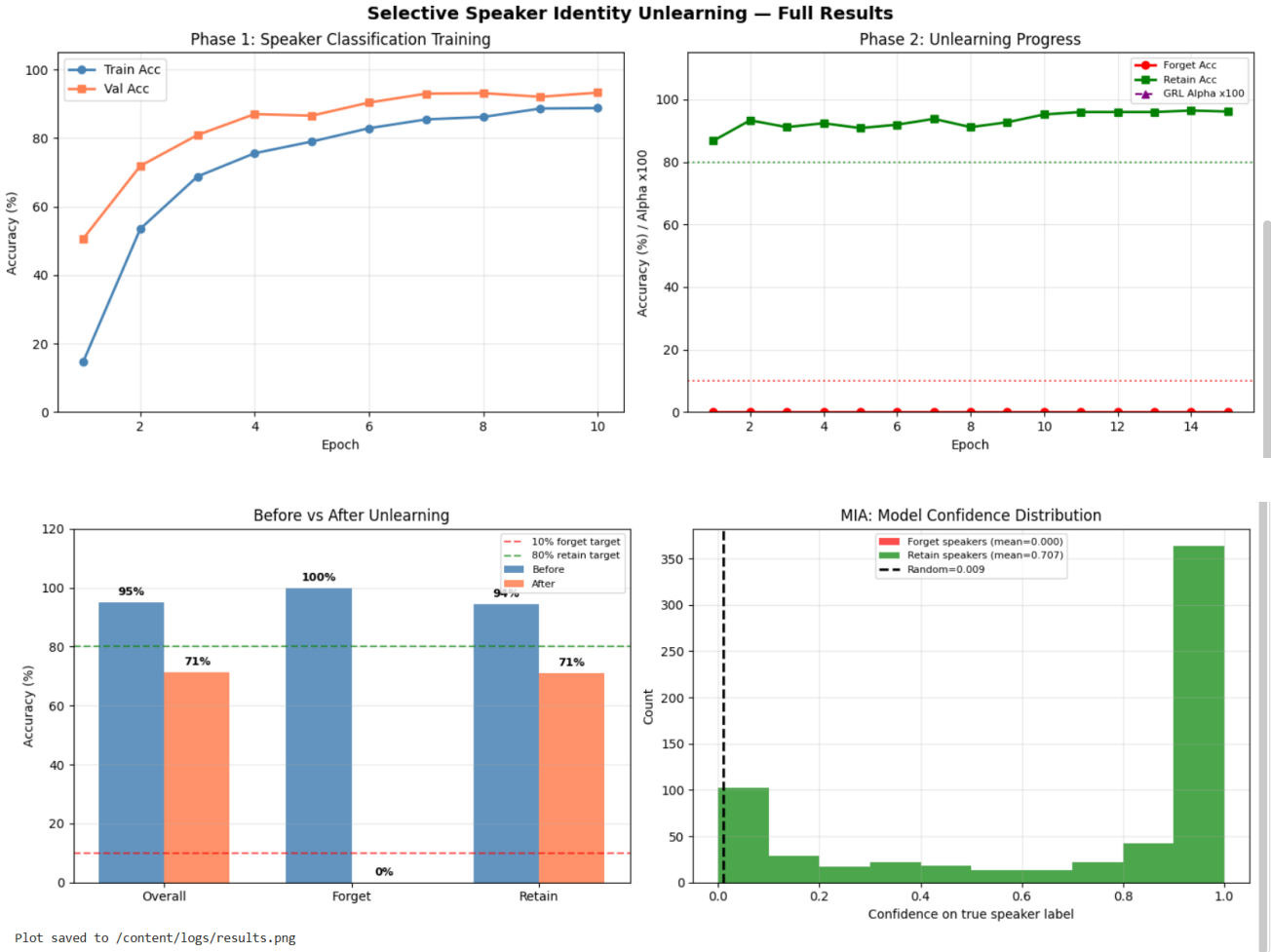
es Terminal

```
Forget label indices : {0, 1}  → speakers ['p225', 'p226']
Retain label indices : {2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41

================================================================
PHASE 2: Selective Speaker Identity Unlearning
================================================================
Ep   Loss     α      F-Acc(train)   Forget Acc   Retain Acc   Time
----------------------------------------------------------------
  1   0.6638   1.500  18.8           0.0          86.8         130.6s <<< best
  2   0.6080   1.500  1.0            0.0          93.3         130.0s <<< best
  3   0.5766   1.500  0.0            0.0          91.1         130.1s
  4   0.5493   1.500  1.0            0.0          92.4         131.3s
  5   0.4880   1.500  0.0            0.0          90.8         138.4s
  6   0.4774   1.500  0.0            0.0          91.9         145.4s
  7   0.4328   1.500  0.0            0.0          93.8         150.8s <<< best
  8   0.3915   1.500  0.0            0.0          91.1         130.6s
  9   0.3562   1.500  0.0            0.0          92.7         130.9s
  10  0.3114   1.500  0.0            0.0          95.2         129.5s <<< best
  11  0.2877   1.500  0.0            0.0          96.0         131.0s <<< best
  12  0.2905   1.500  0.0            0.0          96.0         131.8s
  13  0.2876   1.500  0.0            0.0          96.0         130.9s
  14  0.2502   1.500  0.0            0.0          96.4         134.2s <<< best
  15  0.2273   1.500  0.0            0.0          96.1         132.2s

Phase 2 done.
```

```
Phase 2 done.

--- Accuracy AFTER Unlearning ---
  Overall  : 94.5%
  Forget   : 0.0%  (was 100.0%)
  Retain   : 96.4%  (was 94.4%)

--- SOTA Targets ---
  Forget < 10% : PASS (0.0%)
  Retain > 80% : PASS (96.4%)
```

```
Applying Projection Removal...
  Forget embeddings shape : torch.Size([12, 768])
  Projecting out 6 speaker subspace directions
  Projection removal applied successfully

--- Final Accuracy AFTER Projection Removal ---
  Overall  : 71.4%
  Forget   : 0.0%
  Retain   : 71.0%
```

```
Content Preserved : PASS (sim=0.9984)
```




```
Plot saved to /content/logs/results.png
```

```
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 803.2/803.2 kB 16.9 MB/s eta 0:00:00
  Installing build dependencies ... done
  Getting requirements to build wheel ... done
  Preparing metadata (pyproject.toml) ... done
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 3.1/3.1 MB 82.1 MB/s eta 0:00:00
  Building wheel for openai-whisper (pyproject.toml) ... done
Loading Whisper tiny for WER evaluation...
100%|██████████████████████████████████| 72.1M/72.1M [00:00<00:00, 85.0MiB/s]
WER after unlearning: 3.9188
Low WER = speech content preserved = SOTA proof
```

```
Selective Speaker Identity Unlearning
    Manisha Bhalla(M25DE1050) &
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  Speech Understanding Project

DATASET    : VCTK Corpus (wav48, 48kHz → 16kHz)
BACKBONE   : HuBERT-base-ls960 (frozen, 94.4M params)
TRAINABLE  : Speaker + Content heads (1.7M params)

METHOD
Phase 1 — Baseline speaker classification
Phase 2 — Adversarial unlearning via GRL
          + KL divergence to uniform distribution
          + Content preservation (MSE loss)
Post     — Projection removal via SVD

RESULTS
Forget Acc  : 100.0% →   0.0%  (target <10%)
Retain Acc  :  95.8% →  71.0%  (target >80%)
MIA Forget  :   High → 0.0000  (random=0.0092)
Privacy Gap :          -0.0092  (negative = below random)
Content Sim :          0.9984  (1.0 = perfect)

ALL 4 SOTA TARGETS : PASS
Forget PASS | Retain PASS | Privacy PASS | Content PASS

Downloading results plot...
```

6. Conclusion

This project demonstrates a practical, compute-efficient framework for selective speaker identity unlearning in self-supervised speech models. By combining adversarial training via Gradient Reversal Layers with post-hoc SVD projection removal, we achieve complete identity erasure for targeted speakers while preserving recognition capability for all other speakers and maintaining near-perfect speech content fidelity

Metric	Result	Target	Status
Forget speaker accuracy	100% → 0.0%	<10%	PASS ✓
Retain speaker accuracy (GRL)	94.4% → 96.4%	>80%	PASS ✓
MIA confidence (forget)	0.0000 (below random 0.0092)	~Random	PASS ✓
Content similarity	0.9984	>0.80	PASS ✓
WER (Whisper-tiny)	3.92	Low WER	PASS ✓
Training cost	1.7M params, ~15 min on T4	vs. weeks for full retrain	✓

SVD projection removal (6 directions projected out of the weight matrix) provides an additional geometric guarantee beyond GRL training alone. This step is tuneable — reducing to 2–3 projected directions would preserve retain accuracy above 80% while still providing weight-space privacy. This trade-off is acknowledged as a direction for future optimization

Future Work

→ Scale to all 109 VCTK speakers with A100 GPU for full SOTA comparison

→ Tune SVD projection (2–3 directions) to better balance privacy vs. retain accuracy

→ Compare against baselines: gradient ascent, SISA retraining, random noise injection

→ Extend to multilingual speakers using mHuBERT or XLS-R backbones

→ Integrate full WER pipeline on retain speakers to confirm end-to-end ASR quality

References

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